Fiscal Year:	FY 2015	**Task Last Updated:**	FY 08/02/2014
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Project Title:	Structure and Response of Spherical Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
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No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
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Flight Assignment:	NOTE: End date changed to 11/24/2014 as new award started 11/25/2014, per discussion with D. Stocker/GRC (Ed., 11/5/15) NOTE: Project extended to 9/30/2015, per D. Stocker/GRC and NSSC, pending award of new grant (Ed., 10/6/14) NOTE (Ed., 8/5/2014): This project is being continued as a new grant to be awarded in fall 2014. See that task for subsequent reporting. NOTE: Project extended to 9/30/2014 per NSSC information/D.Stocker-GRC (Ed., 8/2012) NOTE: Project to be extended to 9/30/2014 per D. Stocker/GRC (Ed., 7/20/2011)		
Key Personnel Changes/Previous PI:			
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Task Description:	The effort was divided into experimental and theoretical elements. The experimental effort consisted of using a porous spherical burner to issue fuel mixtures into an oxidizing environment. Different mixtures of fuel species permitted investigation of the phenomena of interest. The experiment effort was conducted primarily in the drop towers at the GRC and the NASA reduced-gravity aircraft. Space flight experiment was fully defined and approved during the program period. The theoretical effort consisted of the development of computational codes simulating the combustion phenomena of interest, and interpreting the calculated results on the basis of fundamental flame physics. The purpose of this flight definition task is to gain fundamental understanding of the structure and dynamics of diffusion flames in a simple, well-defined flow field. Using a spherical geometry this work considers: 1) gas-phase transient processes, including fuel vapor accumulation inside the flame and far-field diffusion, and their influence on flame movement and the achievement of steady flame behavior; 2) extinction boundaries of steady or near-steady flames and the identification of the associated chemistry at the global, reduced, and detailed levels; 3) the characteristics of partially-premixed flames; 4) flamefront pulsation and cellular instability especially near the extinction limits; 5) the dynamics of rotating flames; and 6) the behavior and structure of quasi-spherical flames in partial gravity environments. These studies are intended to significantly enhance the understanding of diffusion flames, especially providing capability for predicting flame behavior under varied conditions and configurations. As such, this work should contribute to engineering approaches to practical combustion systems, achieving increased energy conversion efficiencies and reductions in unwanted combustion emissions in both terrestrial and extraterrestrial environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced understanding of flame-flow vortical interaction could lead to improved burner design and operation in terms of energy efficiency and pollutant emissions.
	NOTE: This project has been continued as a new grant--Structure and Response of Spherical Diffusion Flames--NNX15AC79A. See that task for subsequent reporting. (Ed., November 2015) FINAL REPORT--During the reporting period results from the following investigations were published. Effects of Soret diffusion on the minimum ignition energy of hydrogen/air mixtures by a heated kernel, and the associated flame structure, are investigated numerically with detailed chemistry and transport. Results show that Soret diffusion leads to larger (smaller) minimum ignition energy for relatively rich (lean) mixtures, and that this effect is mainly engendered by the Soret diffusion of H2, while that of the H radical is almost negligible. Mechanistically, Soret diffusion of H2 increases the fuel concentration entering the ignition kernel and consequently renders rich (lean) mixtures harder (easier) to ignite. The evolvement of the flame structure shows that before the H radical is substantially accumulated to exert an influence on the reaction zone, the high temperature gradient of the ignition initiation has driven H2 in the mixture moving towards the ignition kernel. For the stretched spherical flame of the ignition, it is shown that Soret diffusion of both H and H2 should be included such that the stretched flame speed can be accurately predicted. At small flame radius (large strain rate), the Soret diffusion of H2 is the dominant mode of Soret diffusion; at large flame radius (small strain rate), the influence of Soret diffusion is similar to that on the unstretched adiabatic planar flame where the Soret diffusion of H radical plays the major role. Dynamics & Morphology of Colliding Spherical Flames: A preliminary experimental and computational study is conducted on the dynamics and flame morphology of two identical colliding spherical flames. Experimentally, premixtures of fuel (C2H6, C3H8) and air are ignited by two synchronized sparks in a closed chamber under atmospheric pressure, with the subsequent flame front evolution tracked by a schlieren system with high-speed camera. The images yield three distinctive stages during collision, namely: independent propagation, interface flattening and corner dynamics. The evolution of the total surface area and flame distortion is evaluated. Results show that the Lewis number dependence of the flame propagation speed at the corner is opposite to that expected of the thermal-diffusional effect, thereby demonstrating that Huygens' propagation dominates over the effects of stretch-affected preferential diffusion. Simulation shows that the flow field is largely distorted around the flame corner, causing pressure redistribution and vorticity generation through the baroclinic force across the flame. Weakly Buoyant Spherical Diffusion Flames: Properties of Hydrogen-CO/Ethylene Flames: Flame structure is strongly dependent upon the environment in which the flame is situated, in particular Earth's gravity. In a microgravity or microbuoyancy environment, natural convection due to the unidirectional gravity vector is largely suppressed. As buoyant effects are minimized, spherical symmetry of a fundamentally spherical phenomenon can be achieved. In particular, spherical flames present a unique opportunity for studying flame phenomena, as the spherical symmetry reduces complexity in analysis and interpretation of the controlling physical processes. In this study, weakly buoyant spherical flames are generated using a low pressure and low density fuel environment and a spherical porous copper burner. Flame testing is focused on mixtures of hydrogen and another fuel reacting with diluted oxygen. Ethylene-based fuels are first used, as ethylene is a more complex hydrocarbon than methane and exhibits sooting behavior. Syngas is also explored, for its potential in clean-burning technology. Five fuel mixtures are tested: hydrogen-methane, hydrogen-ethylene, helium-ethylene, nitrogen-ethylene and hydrogen-carbon monoxide. Through observation of flame shape and luminosity, the flames are tested for variations with pressure, oxidizer concentration, fuel concentration, and flow rate. Computational studies are also conducted to elucidate experimental results. This work is reported in: "Weakly Buoyant Spherical Diffusion Flames: Properties of Hydrogen-CO/Ethylene Flames," Laurel Paxton, Princeton University Undergraduate Independent Work. Final Report, May 1, 2014. Role of NTC Chemistry in Flame Phenomena: The phenomenon of negative temperature coefficient (NTC) is an essential feature of the chemical kinetics of large hydrocarbons at low to intermediate temperatures, and is practically relevant to the intrinsically low-temperature ignition processes governing engine knock, accidental explosions, and the recent development of the homogeneous charge compression ignition (HCCI) engines. Furthermore, while fundamental studies of this phenomenon have mostly involved homogeneous systems such as the shock tube, flow reactor, and rapid compression machine, recent microgravity experiments and computational simulation by Dryer and William suggested that the NTC chemistry can also fundamentally affect the diffusion-affected phenomena of heptane droplet combustion. We have initiated a systematic investigation on NTC flames with various flame configurations, such as the counterflow premixed and diffusion flames, the doubly infinite planar premixed flame propagation, and the ignition of a premixture by a heated kernel. The understanding gained from such investigation will complement the microgravity studies of droplet combustion. Self-acceleration of cellular spherical flames: The extent of rigor in the various claims of self-similarity in previous

studies of the self-acceleration of expanding spherical flames is examined, particularly on the attainment of 3/2 for the power-law acceleration exponent. We find that the evidences presented in some of the prior studies were not rigorously interpreted, and as such the claim of the 3/2 exponent, motivated by the initial suggestion of Gostintsev et al. based on results of uncertain accuracy, and by the attractiveness of its potential implication on self-turbulization, is not justified. Our results suggest that self-acceleration definitely exists; and self-similarity, subsequent to the initial transient, most likely exists. Our power-law exponent, however, still seems to weakly depend on the system pressure and the intensity of the diffusional-thermal instability, and as such has not reached the hydrodynamic, flame-sheet limit. Further experiments with even thinner flames and longer observation times are needed to resolve this scientifically and practically important issue on flame propagation.This work is reported in: “An experimental investigation on self -acceleration of cellular spherical flames,” Fujia Wu, Grunde Jomaas and Chung K. Law, Proceedings of the Combustion Institute, 34 (2013) 937–945.

Theoretical analysis of low-Grashof number flames: The weakly-buoyant, quasi-steady burning of a spherical droplet in an infinite medium is studied theoretically for unity Lewis number and infinite chemical reaction rate, using the Boussinesq model. Approximate solution is given to the non-linear outer equations by applying the Oseen linearization to the unbounded low Grashof number (Gr) flow. This outer solution is then matched to the inner convective-diffusive solution treated with the Schvab-Zeldovich formulation. Half-order asymptotic expansions are adopted to describe the temperature and mass-fraction profiles. The flame-sheet position, as a function of polar angle and radial position, and mass burning rate are determined. The results compare favorably with experiments. In our perturbation analysis for small normalized density difference, flame characteristics are found to scale with Gr1/2.

Experiments on spherical flames in low pressure: Complementary low-pressure experiments are being used to examine the dynamics and flame structure of low-buoyancy, inverted, spherical diffusion flames. The effect of buoyancy is minimized in a low-pressure environment, typically down to 0.1 atmosphere, beyond which kinetic effects would become severe. However, since the flames examined have centimeter-order diameters, buoyancy can still be significant due to the large dimension. Therefore, to further reduce buoyancy, an additional arrangement is implemented, which involves injecting the oxidizer gas into an environment of low molecular weight fuel such as hydrogen and methane. Buoyancy is reduced because the hot, low-density flame sphere is now situated in an environment of low molecular weight gas. As such, spherical and rotating non-buoyant diffusion flames can be investigated, for extended durations. Detailed measurements are being made of the flame configuration, as well as its temperature and species (CH and OH) profiles, for different fuels and mixtures involving hydrogen, methane, and ethylene, along with inerts composed of nitrogen, helium, and carbon dioxide.

Transient numerical simulation: The transient numerical simulation of spherical flames developed as part of this program includes detailed chemistry and transport models, especially radiation mechanisms. The numerical simulations are being used as a predictive tool to assess the test matrix of the SRD for the space-based experiments. Specifically, cases of kinetic and radiative extinction are being investigated to assess the corresponding flow rates and times needed to achieve the phenomena. Only the cases which should reveal the phenomena will remain, along with some safety-factor redundancies. Due to time and g-jitter limitations, little of the planned test matrix can be adequately explored in the ground based test facilities, and the ACME definition will rely on the best effort in the modeling work. The experiments are meant to examine the relative importance of radiative emission and re-absorption, the adequacy of the detailed chemical kinetics mechanism and transport sub-models used, and the onset of flame front instabilities that could modify the extinction boundaries. Constant pressure and constant volume cases are also being compared.

Task Progress:

Spherical diffusion flames under rotation: Microgravity experiments are conducted in the 2.2-second drop tower and zero-gravity facility at NASA-GRC to gain fundamental understanding of the effects of spinning on an otherwise spherical diffusion flame. The flames are generated by injecting either a fuel or an oxidizer mixture from a porous burner to a controlled ambient of either an oxidizer or fuel mixture, respectively. Results show that the polar flame location scales with the angular velocity monotonically as where a is greater and smaller than unity for small and large spinning velocities, respectively. On the contrary, the equatorial flame location responds nonmonotonically to increasing spinning velocity: first increasing and then decreasing. The experimental observations agree well with the computational simulation where the simulated results demonstrate that the nonmonotonic response of the equatorial flame location is caused by dilution of the reactant concentration in the outwardly-directed radial flow by the product and inert that are carried by the inwardly-directed polar flow upon traversing the flame segment in the polar region. This work was reported in: “Response of spherical diffusion flames subjected to rotation: microgravity experimentation and computational simulation,” by S. W. Yoo, S. Chaudhuri, K. R. Sacksteder, P. Zhang, D. L. Zhu and C. K. Law, Combustion and Flame, Vol. 159, pp. 665-672 (2012). (See FY2012 report Bibliography.)

Soret diffusion in heptane flames: The influence of Soret diffusion on the structure and response of n-heptane/air flames is investigated numerically with detailed reaction mechanism and transport. Unstretched freely-propagating planar premixed flames as well as stretched counterflow premixed and diffusion flames are studied, with emphasis on the separate and combined Soret effects of heptane and its oppositely-directed decomposition species H2 and H, as well as those of the major species O2, N2, CO2 and H2O. Results show that, in the unstretched case for which the flame temperature remains at its adiabatic value, Soret diffusion primarily affects the chemical kinetics embedded in the flame structure and the net effect is small; while in the stretched cases, its impact is mainly through those of heptane and the secondary fuel H2 in modifying the flame temperature, with substantial effects, while H2 and H also affect the chemical kinetics especially when the active reaction zone is localized. The dilution/enrichment of the reactant concentrations in the flame through the Soret diffusion of the major species O2, N2, CO2 and H2O further exert finite effects on the flame burning intensity. Self-acceleration of expanding spherical flames induced by flamefront instability: The cells that continuously develop over the flame surface of an expanding spherical flame increase its area and thereby the global propagation rate, resulting in the possibility of self-acceleration. The present study examines whether this self-acceleration could be self-similar, and, if so, whether it could also be self-turbulizing. Extensive experiments at elevated pressures and thereby reduced laminar flame thicknesses and enhanced propensity to exhibit Darrieus-Landau instability were conducted for hydrogen/air over an extensive range of equivalence ratios. The results demonstrate the strong possibility of self-similar flame acceleration, strong influence of the system pressure, moderate influence of diffusional-thermal instability, and a corresponding moderate spread in the power-law acceleration exponent.

Review on future fuel options: The state of research on developing fuel options for next generation chemical propulsion is reviewed for aviation fuels and energetic fuels. For aviation fuels, the development is based on considerations of cost, energy security and climate change, with Fischer-Tropsch synthetic fuels and biofuels hold potential as alternative aviation fuels. The need for basic research to develop predictive capability for the oxidative chemistry of evolving fuels in evolving engine designs is emphasized, illustrated by the intricate reaction pathways and the enormity of the reaction

mechanisms involved. Recent research activities towards achieving the goal of fuel design are discussed through the development of detailed mechanisms, reduced mechanisms, and surrogate fuels. For the development of high-energy-density propellants, advances in several classes of materials are discussed, including metallized and hypergolic propellants, and propellants with strained and functionalized molecules as well as nanoparticle addition. The impact of the recent progress in chemical synthesis, materials science and nano science on these advances is noted. This work is reported in: “Fuel options for next generation chemical propulsion,” by C.K. Law, AIAA Journal, AIAA Journal, Vol. 50, No. 1, pp. 19-36 (2012). doi: 10.2514/1.J051328. (Reported in FY2012 Bibliography.)

Accelerating deflagration front: The formulation of Taylor on the self-similar propagation of an expanding spherical piston with constant velocity was extended to an instability-wrinkled deflagration front undergoing acceleration with RF (infinity) ta, where RF is the instantaneous flame radius, t the time, and alpha a constant exponent. The formulation describes radial compression waves pushed by the front, trajectories of gas particles, and the explosion condition in the gas upstream of the front. The instant and position of explosion are determined for a given reaction mechanism. For a step-function induction time, analytic formulas for the explosion time and position are derived, showing their dependence on the reaction and flow parameters including thermal expansion, specific heat ratio, and acceleration of the front.

Transient behavior of spherical diffusion flames: The transient behavior of burner-supported spherical diffusion flames was studied in the transport-induced limit of low mass flow rate and the radiation-induced limit of high mass flow rate which characterize the isola response of flame extinction. Oscillatory instability was observed near both steady-state extinction limits. The oscillation typically grows in amplitude until it becomes large enough to extinguish the flame. The oscillatory behavior was numerically observed using detailed chemistry and transport for methane (50%CH4/50%He into 21%O2/79%He) and hydrogen (100% H2 into 21%O2/79%He) diffusion flames where the fuel was issued from a point source, and helium was selected as an inert to increase the Lewis number, facilitating the onset of oscillation. In both methane and hydrogen flames, the oscillation always leads to extinction, and no limit cycle behavior was found. The growth rate of the oscillation was found to be slow enough under certain conditions to allow the flame to oscillate for over 450 seconds, suggesting that such oscillations can possibly be observed experimentally. For the hydrogen flames, however, the frequency of oscillation near the transport-induced limit is much larger, approximately 60 Hz as compared to 0.35 Hz for the methane flame, and the maximum amplitude of temperature oscillations was about 5K. The distinctively different structures of the hydrogen and methane flames suggest that while both instabilities are thermal-diffusive in origin, oscillations in the hydrogen flames resemble those of premixed flames, while oscillations in the methane flames are non-premixed in character.

Structure and extinction characteristics of counterflow diffusion flames: The structure and extinction characteristics of counterflow diffusion flames with flame radiation and non-unity Lewis numbers of the fuel and oxidant were examined using multi-scale asymptotic theory and a model expressed in terms of the jump relations and reactant leakages with the proper consideration of the excess enthalpy overlooked in previous analyses is developed. The existence of the dual extinction limits in the presence of radiative heat loss, namely the kinetic limit at small Damkohler number (high stretch rate) and the radiative limit at large Damkohler number (low stretch rate), are identified. It is found that the former one is minimally affected by radiative loss, while substantial amount of heat loss is associated with the radiative limit. The influence of radiative loss on the extinction Damkohler numbers is found to be through its effects on the flame temperature, excess enthalpy and the reduced extinction Damkohler number. At both extinction limits, the contribution from the flame temperature is always important and dominant. The contributions from the other two, however, could be important in some special cases. At small Le (fuel), the contribution from the reduced extinction Damkohler number is large and even dominant under small radiative loss. The contribution from the excess enthalpy is important for small Le (oxidizer) and it may be comparable to the contribution from the flame temperature when radiative loss is small. Thus, overlooking the excess enthalpy by previous analyses may result in rather large error in the predicted extinction Damkohler numbers, especially the kinetic one. Onset of flame oscillation: Using the results derived from a previous study, a linear stability analysis was conducted to study the onset of near-limit flame oscillation with radiative heat loss in 1-D chambered planar flames using multi-scale activation-energy asymptotics. The oscillatory instability near the radiation-induced extinction limit at large Damkohler numbers is identified, in additional to the one near the kinetic limit at small Damkohler numbers. It is shown that radiative loss assumes a similar role as varying the thermal diffusivity of the reactants. Thus, flame oscillation near the radiative limit is still thermal-diffusive in nature although it may develop under unity Lewis numbers. The unstable range of Damkohler numbers near the radiative limit shows quite similar parametric dependence on the Lewis numbers of reactants, Le (fuel) and Le (oxidizer), the equivalence ratio, and the radiative loss as that near the kinetic limit. They both increase monotonically with Le (oxidizer) and equivalence ratio and increase then decrease with Le (fuel). Increasing radiative loss extends the parameter range under which flame oscillations may develop. However, they show different dependence on the temperature difference between the supplying reactants. Unless radiative loss approaches its maximum value the system can sustain, flame oscillation near the radiative limit is only possible within a limited range of delta-T, whereas it is promoted monotonically with decreasing delta-T near the kinetic limit. Furthermore, while radiative loss shows small effect on the nondimensional oscillation frequency, the dimensional frequency of flame oscillations near the radiative limit can be substantially smaller than that near the kinetic limit.

Bibliography Type:	Description: (Last Updated: 11/26/2025)
Dissertations and Theses	Paxton L. "Weakly Buoyant Spherical Diffusion Flames: Properties of Hydrogen-CO/Ethylene Flames." Princeton University Undergraduate Independent Work. Final Report, May 1, 2014. , May-2014
Papers from Meeting Proceedings	Yang S, Chaudhuri S, Zhu D, Law CK. "Dynamics and Morphology of Colliding Spherical Flames." Fall Technical Meeting of the Eastern States Section of the Combustion Institute 2013, Clemson, SC, October 13-16, 2013. In: Fall Technical Meeting of the Eastern States Section of the Combustion Institute 2013. Proceedings of a meeting held 13-16 October 2013, Clemson, South Carolina, USA. Pittsburgh, Pa. : The Combustion Institute, 2014. Printed by Curran Associates, Inc., ISBN: 9781629937199. , Apr-2014